Prabhavit Gupta



Generative AI Engineer: Enterprise GenAI Solutions, Agentic Workflows & Production Systems

+39 351 610 7807 | prabhavitg@gmail.com | linkedin.com/in/prabhavit | github.com/prab-gupta | Venice, Italy (Available for Milan / Hybrid)

Summary

Generative AI Engineer with 3 years of hands-on production experience architecting enterprise GenAI applications, autonomous agent workflows, and scalable integration systems. Proven track record partnering directly with stakeholders to identify critical operational challenges, define pragmatic AI roadmaps, and engineer hardened, observable solutions from concept to live deployment. Technical foundation spanning Python (FastAPI, backend services), TypeScript/JavaScript (Node.js, Next.js), PostgreSQL, and cloud environments (AWS/GCP), with deep practical specialization in applied generative AI: multi-step orchestration, tool and function calling, vector stores and semantic search, prompt/context engineering, and automated evaluation loops over live operational logs. Pragmatic builder who operates daily with AI-native coding tools (Claude Code, Cursor) and designs architectures that deliver measurable business impact. Italian resident based in Venice with EU Long-Term Residence Permit (permanent unrestricted Italian/EU work authorization without visa sponsorship); available for Milan hybrid collaboration.

Technical Expertise

- **Generative AI & Agent Architectures:** Large language models, multi-step agent orchestration, tool and function calling, schema-enforced JSON validation, guardrails, fallback escalation, context window optimization.
- **Search, Retrieval & Vector Stores:** Retrieval-Augmented Generation (RAG), vector embeddings, vector databases, hybrid keyword/semantic search, document ingestion pipelines, knowledge grounding against live databases.
- **Production LLMOps & Evals:** Automated evaluation pipelines analyzing real production conversation logs, hallucination prevention, latency profiling, token economics, cost optimization, trace observability.
- **Full-Stack & Backend Engineering:** Python (FastAPI, microservices), TypeScript / JavaScript (Node.js, Next.js, React), RESTful APIs, webhook synchronization, automated contract and regression testing (Playwright).
- **Data Infrastructure & DevOps:** PostgreSQL, relational schema design, Supabase, Docker containers, cloud deployments (AWS/GCP), CI/CD pipelines, Git, workflow automation (n8n).
- **Work Authorization & Languages:** English (C1 Professional), Hindi (Native), Italian (Beginner). EU Long-Term Residence Permit (permanent unrestricted work authorization in Italy, no visa sponsorship required).

Experience

FlowSpace.co **Mar 2024 - Aug 2026**
Lead Developer (Full-time, Freelance) *London, UK (remote)*

- Architected, deployed, and maintained the production AI customer support agent runtime on a live platform serving 4,500+ active users, automating customer inquiries across pricing, venue inventory, and bookings.
- Grounded agent responses against live operational databases and partner APIs (OfficeRnD, Nexudus) using strict JSON tool schemas and fallback escalation logic that routed complex inquiries to staff.
- Built automated evaluation loops analyzing real customer conversation transcripts rather than synthetic benchmarks, systematically detecting failure patterns and eliminating hallucinated answers.
- Profiled latency and API token consumption across model versions, optimizing prompt templates and context window utilization to reduce inference costs while maintaining rapid response times.
- Deployed automated end-to-end regression test suites using Playwright, verifying API contracts and continuous deployment stability across end-to-end customer booking flows.
- Spearheaded platform migration from legacy architecture to a modern Next.js and backend service stack, supporting platform growth from 2,000 to 4,500+ users without operational degradation.
- Built automated data ingestion and transformation pipelines parsing and validating unstructured partner listings, expanding platform inventory to over 900 locations.

Urlink **Apr 2025 - Jul 2025**
Project Manager & AI Assisted Developer (Freelance) *Barcelona, Spain (remote)*

- Architected core web application backend workflows, defining clear API contracts, relational schemas, and data transformation logic for client-facing features.
- Implemented automated validation and synchronization pipelines connecting third-party services with PostgreSQL, preventing malformed inputs and ensuring relational data integrity.

- Integrated automated CI/CD workflows and regression checks to accelerate feature delivery across distributed development environments.

Independent Technical Consultant **Aug 2023 - Feb 2024**
AI Solutions & Automation Engineer *Venice, Italy*

- Designed and delivered bespoke AI automation workflows, web extraction engines, and API integrations for European SME clients using Python, TypeScript, and Playwright.
- Translated manual, fragmented business operations into automated software solutions with structured logging, retry queues, and reliable recovery logic.
- Engineered webhook listeners and real-time synchronization pipelines across CRM platforms, operational tools, and relational datastores.

Education

Ca' Foscari University of Venice **2020 - 2024**
BA in Philosophy, International Studies and Economics *Venice, Italy*

- Rigorous grounding in formal logic, analytical problem solving, systems analysis, and macroeconomic decision models.